

OUTPUTS

Part 1:

```
=== Baseline Linear Regression ===  
Coefficients: [ 4.48674910e-01  9.72425752e-03 -1.23323343e-01  7.83144907e-01  
-2.02962058e-06 -3.52631849e-03 -4.19792487e-01 -4.33708065e-01]  
Intercept: -37.02327770606391
```

```
Train MSE: 0.5179331255246699  
Test MSE: 0.5558915986952422
```



```
=== Hyperparameter Tuning Results ===  
Best Ridge alpha: 1.0  
Best Ridge CV MSE: 0.5192628644116295  
Best Lasso alpha: 0.001  
Best Lasso CV MSE: 0.519274809485791
```

```
=== Ridge (L2) with best alpha ===  
Coefficients: [ 4.48510924e-01  9.72596535e-03 -1.23014157e-01  7.81416761e-01  
-2.02581346e-06 -3.52585878e-03 -4.19786908e-01 -4.33680793e-01]  
Train MSE: 0.5179332149226819  
Test MSE: 0.5558034669932196
```

```
=== Lasso (L1) with best alpha ===  
Coefficients: [ 4.44698243e-01  9.78271615e-03 -1.15737191e-01  7.41172257e-01  
-1.85738492e-06 -3.50964130e-03 -4.18521643e-01 -4.31828076e-01]  
Train MSE: 0.5179897619159226  
Test MSE: 0.5538940157172418
```

Part 2:

Baseline coefficients:

```
[[ 1.0274368  0.22145051 -0.36213488  0.0254667 -0.15623532 -0.23771256
 -0.53255786 -0.28369224 -0.22668189 -0.03649446 -0.09710208  1.3705667
 -0.18140942 -0.08719575 -0.02245523  0.04736092 -0.04294784 -0.03240188
 -0.03473732  0.01160522  0.11165329 -0.50887722 -0.01555395 -0.016857
 -0.30773117 -0.77270908 -1.42859535 -0.51092923 -0.74689363 -0.10094404]]
```

Baseline train acc: 0.9582417582417583

Baseline test acc: 0.956140350877193

Best params: {'C': np.float64(100.0), 'penalty': 'l1', 'solver': 'liblinear'}

Best CV acc: 0.9670329670329672

Test acc (best model): 0.9824561403508771

L1 coefficients:

```
[[ 7.42687272e-01 -1.07271803e-01  9.19915844e-02 -1.81557002e-03
 0.00000000e+00  4.75592614e+01 -1.17521126e+01 -1.36161896e+02
 1.98781559e+01  0.00000000e+00  0.00000000e+00  1.72402940e+00
 0.00000000e+00 -1.97655047e-01  0.00000000e+00  0.00000000e+00
 4.98785934e+01  0.00000000e+00  1.84587503e+01  0.00000000e+00
 1.75923231e-01 -4.38783116e-01  5.67396799e-02 -2.06681031e-02
 -2.18096887e+01  7.99446887e+00 -1.46827551e+01 -2.51423008e+01
 -2.53244506e+01  0.00000000e+00]]
```

L1 zero coeffs: 9

L1 train acc: 0.989010989010989

L1 test acc: 0.9824561403508771

L2 coefficients:

```
[[ 5.40272741  0.26573732 -0.52651203 -0.02095479 -2.29922151 -0.2169419
 -3.56980218 -5.0110607 -2.26418385  0.36762747 -0.58036525  3.84169255
 -0.63875637 -0.10714305 -0.40591246  3.57237002  4.36577514 -0.26636925
  0.39354765  0.62630698 -0.20718582 -0.68999884  0.17661962 -0.01813087
 -4.67716843 -0.0140846 -4.46493229 -7.61271125 -6.83571871  0.57463274]]
```

L2 zero coeffs: 0

L2 train acc: 0.9692307692307692

L2 test acc: 0.956140350877193